

Utkarsh Sonawane

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EDUCATION

The Pennsylvania State University

B.S. Computer Science | Capital Honors Program

– GPA: 3.53/4.00 | Dean's List

– Coursework: Data Structures & Algorithms; Object-Oriented Programming and Calculus III (in progress)

Harrisburg, PA

Expected Dec 2028

TECHNICAL SKILLS

Languages: Python, Swift, TypeScript, JavaScript, SQL, Java

Backend & Data: FastAPI, Pydantic, PostgreSQL (row-level security), Supabase, Node.js, Express, REST APIs

Apple & Web: SwiftUI, HealthKit, WatchConnectivity, Apple Foundation Models, React, Vite, PDF.js

Testing & Tools: pytest, XCTest, Vitest, mypy, GitHub Actions, Git, Linux

PROJECTS

Nytr — Nutrition & Training Planner | Python, FastAPI, PostgreSQL, SwiftUI

2026

- Built an iOS app and FastAPI backend for deterministic meal planning from Penn State Harrisburg dining menus, integrating HealthKit, Hevy training history, and barcode-scanned foods.
- Designed immutable, provenance-tracked data with versioned foods and menus, append-only corrections, Decimal arithmetic, and PostgreSQL row-level security.
- Kept on-device AI downstream of computed results; 1,135 backend tests (1,124 passing in CI against Postgres) under strict mypy, plus 251 iOS XCTest cases.

Talks — Apple Watch Meeting Recorder | Swift, SwiftUI, watchOS

2026

- Built a watchOS/iOS app that records on the Watch, transfers audio via WatchConnectivity, transcribes on device, summarizes with Apple Foundation Models, and publishes to Notion.
- Implemented ACK-gated audio deletion, a persistent job queue surviving app termination, and idempotent Notion retries; 57 XCTest cases passing in CI.

Devvy — Local Discord Presence Daemon | JavaScript, Node.js, Discord IPC

2026

- Built a Node.js daemon that merges developer-tool activity into one Discord presence using priority arbitration, source expiry, binary IPC framing, and reconnect with backoff.
- Enforced a privacy boundary excluding prompts, code, and file paths; maintained 26 CI test scripts, including a mock Discord IPC server.

MedClarity — Medical PDF Q&A | TypeScript, React, PDF.js, Gemini

2026

- Built a React app for medical PDF Q&A with browser-side text extraction and page citations validated against loaded documents before linking.
- Protected the serverless proxy with size limits, per-client rate limiting, and HTML escaping; 66 Vitest unit tests passing in CI.

RESEARCH EXPERIENCE

Penn State

Undergraduate Researcher — LLM-Based Adverse Drug Reaction Mining

Harrisburg, PA

Oct 2026 – Present

- Researching the detection of adverse drug reactions (ADRs) and extraction of suspected drug-reaction relationships from unstructured Reddit posts and comments using LLMs.
- Research plan includes human-annotated data and evaluation of direct prompting, retrieval-augmented generation (RAG), and agent-based verification using precision, recall, F1, and error analysis.

LEADERSHIP & ACTIVITIES

IEEE Student Branch, Penn State Harrisburg

Technical Chair (Sep 2026 – Present); Technical Team Member (Sep 2025 – Sep 2026)

Harrisburg, PA

Sep 2025 – Present

- Elected Technical Chair in Sep 2026; coordinate software, AI, and robotics workshops and projects for the branch.
- Hosted 2 hands-on workshops on Markdown and building with Google Gemini.

HONORS & CERTIFICATIONS

USAII Global AI Hackathon 2026: Finalist team (5 members) among 6,081+ participants; built [ScholarAI's data pipeline](#) (normalization, embeddings, pgvector).

Certifications: Machine Learning Specialization (*DeepLearning.AI*) | Practical Deep Learning for Coders (*fast.ai*) | Ethics of AI (*University of Helsinki*) | AI Essentials (*Penn State*)